इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते

SwaYaan

DRONE FEDERATION INDIA (DFI)

NIDAR

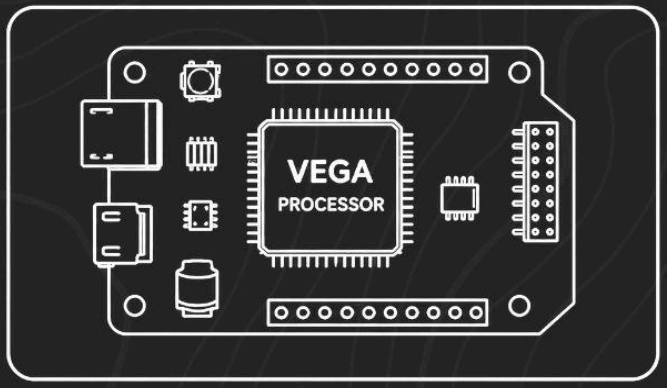
NATIONAL INNOVATION CHALLENGE FOR DRONE APPLICATION AND RESEARCH

Building the Next Generation of Drone Innovators

RULEBOOK | 2026-27







www.nidar.org.in

Version: 2.0

About SwaYaan

SwaYaan, a national initiative by the Ministry of Electronics and Information Technology (MeitY), is dedicated to strengthening India's UAS/Drone ecosystem, aligning with the Government's vision to establish India as a global drone hub by 2030. The initiative empowers learners from undergraduates to faculty and opens learners across five key technical areas, conducting over 1,500 academic, research, and knowledge-sharing activities. Implemented through a network of 30 premier academic and R&D institutions, SwaYaan collaborates with skill councils and industry bodies to drive technological advancements, meet market demands, and set industry standards, ensuring a transformative impact on drone innovation.

About Drone Federation India

Drone Federation India (DFI) is a non-government, non-profit industry body representing over 550+ drone companies and 5,500+ drone pilots in the country, with a mission to promote the design, development, manufacturing, and export of Indigenous drone and counter-drone solutions in the country. DFI plays a pivotal role in policy advocacy, trade promotion, skill development, standards development, and community building by bringing together government, industry, academia, and civil society. DFI hosts several events, like Bharat Drone Mahotsav, which was inaugurated by the Hon'ble Prime Minister Shri Narendra Modi, and Bharat Drone Shakti, which was inaugurated by Hon'ble Defence Minister Shri Rajnath Singh. DFI drives India's leadership in the global landscape through its global alliances and collaborations.

About NIDAR

The National Innovation Challenge for Drone Application and Research (NIDAR) is India's largest and most impactful drone innovation challenge, designed to push the boundaries of engineering, innovation, and real-world problem-solving. It fosters cross-functional collaboration and focuses on addressing critical challenges in drone technology, particularly in disaster management. By engaging students with problem statements that reflect real-world needs, NIDAR encourages industry-oriented research, driving innovative solutions such as geotagging survivors and delivering survival kits using drones.

Beyond technical problem-solving, NIDAR provides participants with invaluable industry exposure through mentorship, guidance, and internship opportunities. This experience extends beyond engineering, equipping students with insights into business practices and entrepreneurship, helping them navigate the path to success in both technology and industry leadership. With its strong emphasis on applied research and industry collaboration, NIDAR aspires to develop the next generation of leaders in the drone sector, empowering them to drive the future of aerospace innovation.

Record of Revision

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Table of Contents

1. About NIDAR **1**

NIDAR 2025-26 1

Key Objectives of NIDAR 1

2. About the 2nd Edition **2**

NIDAR 2026-27 2

Track 1 - Drone Innovation 2

Track 2 - Component Innovation 2

Mission 1 - NIDAR RescueSwarm 2

Mission 2 - NIDAR AirMouse 3

Mission 3 - NIDAR VEGAPilot 3

Detailed Mission Briefing 3

3. Team Eligibility **4**

General Eligibility 4

Track 1 – Drone Innovation 4

Track 2 – Component Innovation 4

4. Competition Structure **5**

Phase 1 - Application & Registration Fees 5

Phase 2A - Verification Process 6

Phase 2B - Online Presentation & Shortlisting 6

Phase 3A & 3B - Progress Review Round 1 & 2 6

Phase 4A - Design Review Presentation 6

Phase 4B - Business Strategy Presentation 7

Phase 4C - Pre-Flight Inspection 7

Phase 4D - Final Mission 8

Phase 5 - NIDAR Awards 8

5. Competition Timeline **9**

6. Awards & Recognition **10**

7. Deliverables **10**

8. General Technical and Operational Requirements **11**

General Rules..... 11
Rules for M1 - NIDAR RescueSwarm..... 11
Rules for M2 - NIDAR AirMouse..... 12
Rules for M3 - NIDAR VEGAPilot..... 13
9. Scoring Structure..... 15
Overall Scoring Criteria..... 15
Mission 1 - NIDAR RescueSwarm..... 15
Mission 2 - NIDAR AirMouse..... 18
Mission 3 - NIDAR VEGAPilot..... 20
Tie Breaking..... 23
10. Code of Conduct & Dispute Resolution..... 24
Professional Conduct..... 24
Safety Compliance..... 24
Academic Integrity..... 24
Organiser Authority..... 25
Dispute Resolution..... 25

Annexure 1: Mission Brief - NIDAR RescueSwarm

Annexure 2: Mission Brief - NIDAR AirMouse

Annexure 3: Mission Brief - NIDAR VEGAPilot

1. About NIDAR

NIDAR 2025-26

The National Innovation Challenge for Drone Application and Research (NIDAR 2025-26), organised by the Drone Federation India in collaboration with the Ministry of Electronics and Information Technology under Project SwaYaan, emerged as one of India's largest student-focused drone innovation challenges. The previous edition was launched by the Secretary, MeitY, in the presence of the Chairman, AICTE, and leading industry stakeholders, and witnessed participation from 3,448 students across 22 States and 4 Union Territories, spanning 109 cities. The challenge focused on Disaster Management and Precision Agriculture. It successfully demonstrated the importance of hands-on innovation, indigenous capability development, and real-world problem-solving through drones and other emerging technologies.

Key Objectives of NIDAR

- To create a national innovation platform for student-led research and development in autonomous drone systems and indigenous avionics.
- To promote innovation in collaborative drone systems, GPS-denied navigation, indoor autonomy, and mission planning technologies.
- To encourage indigenous design and development of drone flight controllers, processors, and critical electronic subsystems.
- To strengthen collaboration between academia, government, defence stakeholders, and industry.
- To provide students with hands-on exposure to real-world operational and strategic drone challenges.
- To support India's vision of technological self-reliance and indigenous capability development in emerging technology sectors.

2. About the 2nd Edition

NIDAR 2026-27

Building on the success and national response to the first edition, NIDAR 2026-27 is envisioned as a larger national innovation platform to advance autonomous drone systems, indigenous avionics, autonomous navigation, and critical drone components. With the rapid growth of India's drone ecosystem and increasing strategic importance of autonomous systems, there is a strong need to encourage student-led innovation in areas such as GPS-denied navigation, collaborative drone operations, indigenous flight-control systems, and semiconductor-linked drone technologies. NIDAR 2026-27 seeks to bridge the gap between academia, industry, and government by providing students with opportunities to solve real-world operational challenges through applied innovation. This year's edition is divided into two tracks:

Track 1 - Drone Innovation

Track 1 focuses on the design, development and deployment of complete drone-based solutions for real-world challenges. Participants are required to integrate hardware, software, autonomy, sensing, communication and mission-planning capabilities to demonstrate a fully functional, end-to-end drone system. Track 1 has two missions.

Track 2 - Component Innovation

Track 2 focuses on the design, development and validation of critical drone subsystems and enabling technologies for unmanned aerial platforms. Participants are required to integrate hardware, software, sensors and control algorithms to develop and demonstrate a functional autopilot using the Aries v3 development board. Track 2 has 1 mission.

Mission 1 - NIDAR RescueSwarm

A severe flash flood has affected a semi-urban settlement located near a river basin. Roads are blocked, several houses are inaccessible, and the local rescue team has received unverified reports of stranded people across a large area of the affected zone. Telecom towers in the affected region are down, mobile connectivity is unavailable, and ground teams cannot rely on external communication networks for real-time coordination. Ground access is delayed, and the first few minutes of response are critical for locating survivors and delivering basic medical support.

The rescue command centre has identified a 10-hectare search area where up to 10 survivors may be present. Since the area is too large for a single drone to cover efficiently, innovation teams must deploy a collaborative autonomous multi-drone system that can rapidly search the area, identify survivors, geotag their locations, and deliver a survivor kit to each identified survivor.

The goal of the innovation teams is to find survivors and deliver a survivor's kit to them as quickly as possible, while ensuring that all drones operate through a single ground control system, report mission data to a single operator, and complete the mission with minimal human intervention.

Mission 2 - NIDAR AirMouse

A building has been severely affected by an earthquake or structural collapse, making several internal passages and rooms unsafe for immediate entry by rescue personnel due to debris, damaged structures, low visibility and the risk of further collapse. Initial reports indicate that survivors may be trapped inside, but their exact locations are unknown.

Since GPS signals are unavailable inside the building, rescue teams require an autonomous aerial system capable of entering the structure, navigating enclosed corridors and rooms, detecting survivors, generating a 2D map of the explored area and tagging survivor locations on the map.

Innovation teams shall develop an autonomous indoor drone system capable of entering a covered maze-like environment through a designated entry point, navigating passages and room-like sections, locating up to six survivors, generating a 2D map, tagging survivor locations and exiting through the designated exit point.

The objective is to autonomously navigate an unknown GPS-denied indoor environment and provide rescue teams with a usable 2D map showing the location of each detected survivor in the shortest possible time.

Mission 3 - NIDAR VEGAPilot

India's drone ecosystem is expanding across defence, agriculture, logistics, infrastructure, disaster response, surveillance and public service delivery. However, the flight controller or autopilot remains one of the most critical drone subsystems, responsible for flight stability, navigation, communication, sensor integration and mission execution.

India's drone electronics ecosystem continues to depend significantly on imported flight controllers, processors, microcontrollers, integrated circuits and supporting components. Building a truly indigenous drone ecosystem, therefore, requires students and innovators to design, develop, integrate, and demonstrate reliable autopilot systems based on Indian processor platforms and non-Chinese electronic components.

Innovation teams shall develop a functional drone autopilot using the **Aeries v3 development board**. Each selected team shall receive two Aeries v3 development boards for development, testing and final integration. Teams shall design and develop the autopilot hardware and software stack around the board and demonstrate its ability to control a standard multirotor drone platform.

During the final demonstration, teams shall integrate their Aeries v3-based autopilot with a drone assembled using the standard parts list prescribed under this Rulebook. The platform may include an airframe, motors, ESCs, a GNSS module, a battery, a communication module, and other components.

The objective is to develop an indigenous Aeries v3-based autopilot capable of stabilising, controlling and navigating a real multirotor platform while demonstrating reliable integration with sensors, communication systems and mission-planning software.

Detailed Mission Briefing

Detailed Mission Briefings, including the mission objectives, arena specifications, technical requirements, operational conditions and mission-specific rules for each mission, are provided in respective Annexures.

3. Team Eligibility

General Eligibility

3.1. A participant may be a member of only one team and may participate in only one problem statement.

3.2. Multiple teams from the same institution are permitted.

3.3. No team shall share members, drones, flight controllers, processors, hardware, software, or competition deliverables with another team.

Track 1 – Drone Innovation

3.4. The target group shall consist of students enrolled in Diploma, Undergraduate, Postgraduate or Doctoral programmes in engineering, technology, science or related disciplines.

3.5. Students affiliated with research institutions, national laboratories, Centres of Excellence, government R&D organisations, or technical training institutions established, funded, operated, or accredited by the Government of India, any State Government, statutory body, university, or public sector undertaking shall also be eligible to participate, subject to verification by the Organising Team.

3.6. The student team can consist of a **minimum of 4 members and a maximum of 10 members.**

3.7. The student team **should be interdisciplinary** and have at least one member from a different branch of study or discipline to encourage diverse problem-solving approaches.

3.8. Additionally, each team **should have one faculty member** in addition to the students for general guidance and mentorship.

3.9. All participants must remain enrolled with their respective institution or organisation until the conclusion of the **NIDAR Finals.**

Track 2 – Component Innovation

3.10. The target group shall consist of students enrolled in Diploma, Undergraduate, Postgraduate, or Doctoral programmes in engineering, technology, science, or related disciplines, with a focus on electronics, embedded systems, and related domains.

3.11. Students affiliated with research institutions, national laboratories, Centres of Excellence, government R&D organisations, or technical training institutions established, funded, operated, or accredited by the Government of India, any State Government, statutory body, university, or public sector undertaking shall also be eligible to participate, subject to verification by the Organising Team.

3.12. The student team can consist of a **minimum of 2 members and a maximum of 6 members.**

3.13. Additionally, each team should have **one faculty member** in addition to the students for general guidance and mentorship.

3.14. All participants must remain enrolled with their respective institution or organisation until the conclusion of the **NIDAR Finals as per the Competition Timeline in this Rulebook.**

4. Competition Structure

Phases		Mission 1 - NIDAR RescueSwarm	Mission 2 - NIDAR RescueSwarm	Mission 3 - NIDAR VegaPilot
1	Application & Registration	Yes	Yes	Yes
2A	Verification Process	Yes	Yes	Yes
2B	Online Presentation & Shortlisting	**All Teams Auto-Shortlisted**		**Top 100**
Progress Review Rounds (Online)				
3A	Online Progress Review Round 1	Yes	Yes	Yes
3B	Online Progress Review Round 2	Yes	Yes	Yes
Final Mission (Offline)				
4A	Design Review Presentation	Yes	Yes	Yes
4B	Business Strategy Presentation	Yes	Yes	Yes
4C	Pre-Flight Inspection	Yes	Yes	Yes
4D	Final Mission (Flight)	Yes	Yes	Yes
NIDAR Awards				
5	Award & Felicitation Ceremony	Yes	Yes	Yes

Phase 1 - Application & Registration Fees

4.1. Each team must register online via the official competition website - www.nidar.org.in.

4.2. The registration deadline for the competition shall be posted on the official website.

4.3. In the event of an extension of the registration deadline, the same shall be updated on the official website and promptly communicated to all registered members.

4.4. Each team can choose only one mission in their online application form.

4.5. Each team shall submit the following documents in their online application:
- Letter of Approval from Academic Institution (Can be submitted later after registering online)
- Team Details (along with Govt ID Proof & College ID)
- Proof of Payment of Registration Fee

4.6. The Registration Fee for each team shall be INR 5,000 and shall be paid via the online portal.

4.7. Payment shall not be accepted in Cheque, Demand Draft or any other form.

4.8. The Registration Fee will be non-refundable in any case.

Phase 2A - Verification Process

4.9. The organising team shall verify each registration submission to confirm the completeness and adequacy of the information provided by the participating team.

4.10. Every complete application shall be shortlisted without the application of any additional evaluation or selection criteria. A confirmation email shall be sent to each shortlisted team.

4.11. The confirmation email shall constitute the official confirmation of selection for participation in the competition. The confirmation status shall also be reflected on the official website.

Phase 2B - Online Presentation & Shortlisting

4.12. Teams participating in Mission 1 – NIDAR RescueSwarm and Mission 2 – NIDAR AirMouse shall not be subject to any concept proposal shortlisting process. All teams that complete the registration requirements and submit the prescribed documents shall be automatically shortlisted and eligible to participate in the competition.

4.13. Teams participating in Mission 3 – NIDAR VEGAPilot shall submit a Concept Proposal and undergo an online technical evaluation. Based on the evaluation, the top 100 teams shall receive two VEGA processors each and advance to the development stage.

Phase 3A & 3B - Progress Review Round 1 & 2

4.14. Progress Review Rounds 1 and 2 shall be conducted online to assess the development progress of participating teams. Attendance in both rounds shall be mandatory.

4.15. During each round, teams shall present updates on their development status, key milestones achieved, challenges encountered and planned next steps.

4.16. These rounds are intended solely for progress review and guidance. **No team shall be eliminated based on its performance in either round.** Where required, teams may receive feedback and guidance from the organising team, industry experts and mentors.

Phase 4A - Design Review Presentation

4.17. Each team shall deliver a comprehensive design review presentation to the jury in accordance with the prescribed schedule.

4.18. A total of 25 minutes shall be allocated to each team, comprising up to 15 minutes for the presentation and the remainder for questions and discussion.

4.19. The presentation shall, at a minimum, cover the topics specified under **Scoring Structure of this Rulebook**. Teams may also include any additional information they consider relevant to demonstrate their approach to system design, development, integration, and mission execution.

4.20. There shall be no restriction on the number of slides. However, the prescribed presentation time shall be strictly observed.

4.21. The detailed scoring criteria for the Design Review Presentation are provided in the subsequent sections of this Rulebook.

Phase 4B - Business Strategy Presentation

4.22. Each team shall deliver a comprehensive business strategy presentation to the jury in accordance with the prescribed schedule. The presentation shall be structured as a startup pitch, with the team presenting its drone system or solution as a viable business proposition.

4.23. A total of 25 minutes shall be allocated to each team, comprising up to 15 minutes for the presentation and the remainder for questions and discussion.

4.24. The presentation shall, at a minimum, cover the topics specified under the **Scoring Structure of this Rulebook.**

4.25. Each team shall also submit a detailed cost sheet setting out all costs and expenses incurred in designing and developing the complete solution for the competition.

4.26. There shall be no restriction on the number of slides. However, the prescribed presentation time shall be strictly observed.

4.27. Teams may include additional information they consider relevant to demonstrate their approach to market positioning, commercialisation, fundraising, scalability, and future business opportunities.

4.28. The detailed scoring criteria for the Business Strategy Presentation are provided in the subsequent sections of this Rulebook.

Phase 4C - Pre-Flight Inspection

4.29. Each team shall undergo a mandatory Pre-Flight Inspection, conducted by a designated jury member serving as the Flight Inspector, before undertaking the Final Mission.

4.30. The inspection shall verify the safety, readiness and compliance of the drone system, Command and Control Station and associated equipment with the applicable technical and operational requirements.

4.31. A model Pre-Flight Inspection Checklist shall be released separately.

4.32. Only teams that successfully pass the Pre-Flight Inspection shall be permitted to proceed to the Final Mission.

Phase 4D - Final Mission

4.33. Each team shall be permitted to undertake the Final Mission only after successfully completing the Pre-Flight Inspection.

4.34. The team shall position its Command and Control Station, including all associated equipment such as antennas, displays, computers and remote-control devices, within the designated area. A maximum of two team members shall be permitted to operate or supervise the Command and Control Station.

4.35. Each drone shall be positioned within the designated launch or staging area. One team member may be assigned to position and supervise each drone. The same team member may reset, reload or reposition the assigned drone during the mission, where permitted under the relevant mission rules.

4.36. Each team shall be provided a maximum setup time of five minutes. If the team fails to complete the setup within this period, the Final Mission timer may be started at the jury's discretion.

4.37. For **Mission 1 – NIDAR RescueSwarm**, teams shall demonstrate autonomous multi-drone collaboration for survivor detection, geotagging, mission coordination and survival-kit delivery within the designated mission area.

4.38. For **Mission 2 – NIDAR AirMouse**, teams shall demonstrate autonomous navigation within a GPS-denied indoor environment, survivor detection and localisation, map generation, mission reporting and successful exit from the designated mission area.

4.39. For **Mission 3 – NIDAR VEGAPilot**, teams shall demonstrate the functionality of their VEGA processor-based autopilot through integration with the prescribed drone platform, mission execution, telemetry communication, mission planner integration, and safe flight operations.

4.40. The detailed Mission Briefing, including mission-specific objectives, scoring criteria, technical specifications, arena requirements and operational conditions, is provided in the respective Annexure to this Rulebook.

4.41. The maximum time permitted for each Final Mission shall be 30 minutes. Any objective completed or points earned after the expiry of the prescribed time limit shall not be considered for scoring.

Phase 5 - NIDAR Awards

4.42. The NIDAR Awards Ceremony shall be held at the conclusion of the competition to recognise outstanding performance across all missions.

4.43. Winning teams shall be presented with trophies, awards, certificates and other forms of recognition based on the applicable scoring criteria and their overall performance in the competition.

4.44. The organising team reserves the right to introduce additional awards, special recognitions, or category-specific honours as deemed appropriate.

5. Competition Timeline

5.1. All participating teams must note the essential deadlines as stated in this section. Failing to comply with any deadline may result in disqualification from the competition. Organisers shall have the right to extend deadlines for broader participation and accommodate students' academic activities.

5.2. Important deadlines concerning the competition are as follows:

Phases		**Tentative Timeline**
1	Application & Registration Deadline	2nd Week of August 2026
2A	Verification Process	3rd Week of August 2026
2B	Online Presentation & Shortlisting	3rd Week of September 2026
Progress Review Rounds (Online)		
3A	Online Progress Review Round 1	2nd Week of October 2026
3B	Online Progress Review Round 2	2nd Week of December 2026
Final Mission (Offline)		
4A	Design Review Presentation	January 2027
4B	Business Strategy Presentation	
4C	Pre-Flight Inspection	
4D	Final Mission (Flight)	
NIDAR Awards		
5	Award & Felicitation Ceremony	January 2027

6. Awards & Recognition

No.	Award(s)	Mission 1 - NIDAR RescueSwarm	Mission 2 - NIDAR AirMouse	Mission 3 - NIDAR VegaPilot
1	NIDAR Gold Award	INR 8,00,000	INR 8,00,000	INR 4,00,000
2	NIDAR Silver Award	INR 5,50,000	INR 5,50,000	INR 3,00,000
3	NIDAR Bronze Award	INR 4,00,000	INR 4,00,000	INR 2,00,000
4	NIDAR Ignite Prize (x5)	INR 1,00,000	INR 1,00,000	INR 1,00,000
5	NIDAR Design - Gold Award	INR 75,000	INR 75,000	INR 75,000
6	NIDAR Design - Silver Award	INR 50,000	INR 50,000	INR 50,000
7	NIDAR Business - Gold Award	INR 75,000	INR 75,000	INR 75,000
8	NIDAR Business - Silver Award	INR 50,000	INR 50,000	INR 50,000
9	NIDAR Diaries	INR 50,000		
10	NIDAR Participation Certificate	All participating teams and team members		
11	Additional Benefits (to be announced soon)	Incubation support, Cloud Credits, Design and Simulation Software, Internship Opportunities		

7. Deliverables

7.1. Participating teams shall submit the following deliverables at the relevant stages of the competition:

7.2. **Registration Documents:** Completed online application form, approval letter from the academic institution, team details, and proof of payment of the registration fee.

7.3. **Phase 2B – Concept Proposal and Shortlisting:** A comprehensive Concept Proposal presentation for **Mission 3 – NIDAR VEGAPilot**, submitted in PDF or PPT/PPTX format.

7.4. **Phase 4A – Design Review Presentation:** A comprehensive Design Review presentation submitted in PDF or PPT/PPTX format.

7.5. **Phase 4B – Business Strategy Presentation:** A comprehensive Business Strategy presentation submitted in PDF or PPT/PPTX format; and

Bill of Materials and Cost Sheet: A detailed bill of materials and cost sheet specifying all costs and expenses incurred by the team in designing, developing, integrating and testing its solution.

8. General Technical and Operational Requirements

General Rules

8.1. Teams shall independently design, develop, integrate and demonstrate their complete solutions.

8.2. The use of commercially available, market-ready or ready-to-fly complete drone airframes shall not be permitted. Teams may use commercially available components, open-source software and custom-developed hardware or software, unless otherwise specified in the relevant Problem Statement Annexure.

8.3. All systems and operations shall comply with the applicable safety requirements, technical specifications and competition rules.

8.4. During mission execution, the use of GSM, LTE, 5G, public Wi-Fi, internet connectivity, cloud-based services or any other external network shall be prohibited.

8.5. All communication between drones, onboard systems and the Ground Control Station shall operate through locally deployed communication systems. Optical-fibre cables, wired communication links, physical tethers or any other cable connected to a drone during flight shall not be permitted.

8.6. The organising team and jury reserve the right to inspect any hardware, software, firmware, communication system, source configuration or component used by a participating team for the purpose of verifying compliance.

8.7. Unsafe operation, deliberate interference with another team or competition infrastructure, misrepresentation of system capabilities, or violation of the competition rules may result in penalties, suspension from the mission or disqualification.

Rules for M1 - NIDAR RescueSwarm

8.8. Each team shall deploy at least two or more drones operating as a coordinated multi-drone mission system.

8.9. The combined all-up weight of all deployed drones, including batteries, payloads, sensors and communication equipment, shall not exceed 25 kg.

8.10. All drones shall take off from and land within the designated 12 ft × 12 ft launch and landing area.

8.11. The system shall autonomously search the designated mission area and detect up to ten survivors. Each detected survivor shall be autonomously geotagged, and the corresponding location shall be displayed on the Ground Control Station.

8.12. Teams shall autonomously deliver survival kits to detected survivors. Each survival kit shall weigh 200 grams and measure 20 cm × 10 cm × 5 cm. Payload deployment shall be performed autonomously.

8.13. All drones shall operate through a single Ground Control Station and a unified operator interface. Mission data from all drones shall be consolidated and displayed through the same interface.

8.14. The Ground Control Station shall display, at a minimum:
- Mission status
- Live camera feed from each drone
- Position or estimated position of each drone
- Assigned search area or task for each drone
- Detected and geotagged survivor locations
- Survival-kit delivery status
- Communication and system health status
- Consolidated mission progress and reporting

8.15. The system shall demonstrate collaborative mission execution, including area allocation, task distribution, mission coordination and consolidated mission reporting.

8.16. Any manual waypoint modification, flight-path correction, payload-release command, survivor tagging or mission replanning during mission execution shall be treated as manual intervention.

8.17. The system shall not rely on any external communication network for mission execution, inter-drone coordination or data exchange.

8.18. All drones shall remain within the mission boundary specified by the organising team.

8.19. Each drone shall incorporate, at a minimum, Return-to-Home, communication-loss recovery, low-battery fail-safe, geofence protection and mission-abort capabilities.

Rules for M2 - NIDAR AirMouse

8.20. Each team shall deploy an autonomous drone capable of operating within a GPS-denied indoor environment.

8.21. The drone shall autonomously enter and exit the arena through the same designated access point.

8.22. The total maze or arena area shall not exceed 15 m × 15 m. Each corridor shall provide a minimum clear width of 1 m and a minimum vertical clearance of 8 ft. The standard room dimensions shall be 2 m × 2 m.

8.23. The drone shall take off from within the designated 2-ft × 2-ft launch area.

8.24. The drone shall autonomously navigate through corridors, rooms, turns, junctions and obstacles within the indoor arena. GPS- or GNSS-based navigation shall not be used during mission execution.

8.25. The system shall detect and localise up to six survivors positioned within the arena. Each detected survivor shall be autonomously tagged at the corresponding location on the generated map.

8.26. The system shall generate and display a real-time two-dimensional map of the explored environment.

8.27. The Ground Control Station shall display, at a minimum:
- Mission status
- Live camera feed from the drone
- Drone position or estimated position
- Generated map

- Tagged survivor locations
- Mission progress

8.28. Any manual navigation assistance, map editing, localisation correction, survivor tagging or path-planning input during mission execution shall be treated as manual intervention.

8.29. The drone shall be designed for safe operation within confined indoor spaces and shall avoid contact with walls, ceilings, arena structures and obstacles.

8.30. The drone shall be equipped with propeller covers or guards that fully protect the propeller operating area.

8.31. The drone shall incorporate, at a minimum, emergency-stop, mission-abort, communication-loss recovery and low-battery fail-safe capabilities.

Rules for M3 - NIDAR VEGAPilot

8.32. Teams shall develop a functional flight controller and autopilot system using the Aeries v3 development board as the primary processing platform.

8.33. Participation in the development stage shall be limited to teams shortlisted through the Concept Stage. Each shortlisted team shall receive two Aeries v3 development boards for development, integration, testing and final demonstration.

8.34. Teams shall build and integrate a complete multirotor drone using the standard parts list prescribed under this Rulebook. No additional autopilot-related components shall be used unless expressly permitted by the organising team.

8.35. The autopilot shall support, at a minimum:
- Arming and disarming
- Stabilised flight
- Altitude hold
- Position hold
- Manual flight-mode control
- Safe landing

8.36. The autopilot shall interface with the prescribed sensors, GNSS module, electronic speed controllers, motors and communication systems required for flight operations.

8.37. The autopilot shall communicate with a compatible Mission Planner or Ground Control Station to support:
- Vehicle configuration
- Telemetry transmission
- Vehicle monitoring
- Flight-mode display
- Flight-data recording

8.38. The Ground Control Station shall display, at a minimum:
- Vehicle status
- Vehicle location

- Active flight mode
- Battery status
- Relevant flight and telemetry information

8.39. The system shall record flight and telemetry data for post-flight review.

8.40. During the final demonstration, teams shall integrate the Aeries v3-based autopilot with the drone platform assembled using the standard parts list prescribed under this Rulebook.

8.41. The demonstration shall include:
- Successful arming and take-off
- Taking off and landing in a designated launch and recovery area
- Stabilised flight
- Position hold
- Manual navigation through the prescribed flight course or obstacles
- Autonomous navigation through 5 waypoints
- Telemetry communication with the Ground Control Station
- Return to home
- Safe landing

8.42. All components incorporated within the autopilot system shall comply with the indigenous-component requirements prescribed by the organising team. Chinese-origin components shall not be used within the autopilot system.

8.43. Teams shall submit a complete Bill of Materials identifying the make, model, manufacturer, supplier and country of origin of each component used.

8.44. The autopilot shall incorporate fail-safe provisions for:
- Low-battery conditions
- Communication loss
- GNSS failure
- Unstable flight conditions
- Mission abort
- Emergency landing

8.45. Teams shall not modify, damage or permanently alter any standard component provided by the organising team unless expressly permitted.

8.46. The final demonstration shall establish the successful integration of the Aeries v3-based autopilot, stable flight performance, position-hold capability, telemetry communication, flight data logging, and safe operation of the assembled drone platform.

9. Scoring Structure

Overall Scoring Criteria

Final Mission		Maximum Points
4A	Design Review Presentation	200
4B	Business Strategy Presentation	200
4C	Pre-Flight Inspection	Pass or Fail
4D	Final Mission (Flight)	600
	Total	1000

Mission 1 - NIDAR RescueSwarm

Phase 4A - Design Review Presentation		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	10
2	Overall Mission Architecture & System Design	20
3	Technical Approach, Drone Design & Indigenisation Strategy	20
4	Multi-Drone Collaboration & Task Allocation	30
5	Survivor Detection, Geotagging & Accuracy	30
6	Payload Delivery & Drop Accuracy	25
7	Autonomous Mission Execution & Time Efficiency	30
8	Ground Control Station, Mission Map & Multi-Drone Status Reporting	20
9	Fail-Safe Features & Safety	15
	TOTAL	**200**

Phase 4B - Business Strategy Presentation		
No.	**Parameter & Description**	**Max Score**
1	Problem Understanding & Real-World Relevance	30
2	Target Users, Customers & Beneficiaries	20
3	Market Sizing & Deployment Potential	30
4	Business Model & Revenue Approach	30
5	Expenditure Breakdown & Resource Planning	20
6	Funds Raised, Sponsorships & Resource Mobilisation	20
7	Competitive Advantage & Differentiation	20
8	Go-to-Market Strategy & Partnership	20
9	Regulatory, Safety & Adoption Readiness	10
	TOTAL	**200**

Phase 4C - Pre-Flight Inspection		
No.	**Parameter & Description**	**Result**
1	Pre-flight Inspection shall only be a deciding phase, and successful completion of this phase will allow team members to participate in the Final Mission. Teams shall only be permitted to retry this phase once.	Pass or Fail

Phase 4D - Final Mission - Scoring			
No.	**Criteria**	**Scoring Method**	**Max Score**
1	**Survivor Detection & Geotagging**	25 marks for each correctly detected survivor whose location is also correctly geotagged on the GCS / mission map. Maximum 10 survivors.	250
2	**Survivor Kit Delivery Accuracy**	Marks are awarded for each successful survivor kit drop based on the accuracy zone. Maximum 10 successful drops. Delivery in Zone A (within 1 metre): 20 per drop Delivery in Zone B (within 2 metres): 14 per drop Delivery in Zone C (within 3 metres): 8 per drop	200

3	**Multi-Drone Collaborative Mission Execution**	Yes/No. Full 50 marks if two or more drones operate as one coordinated mission system with shared mission execution, common mission logic, and no independent manual control. Zero if not achieved.	50
4	**Single GCS / Mission Planner Interface**	Yes/No. Full 50 marks if all drones, survivor tags, delivery status, mission boundary, and mission progress are visible in a single operator interface. Zero if not achieved.	50
5	**Fast Completion Bonus**	Yes/No. Full 50 marks if the mission is completed within half of the maximum permitted mission time. Zero if not achieved.	50
	TOTAL		**600**

Phase 4D - Final Mission - Penalties			
No.	**Penalty Condition**	**Penalty Description**	**Max Score**
1	**Landing outside the designated launch or landing zone**	For each drone that lands outside the designated landing zone after mission completion, except in an organiser-approved emergency landing.	-10 per drone
2	**Geofence / Mission Boundary Breach**	For each instance where any drone crosses outside the assigned mission boundary/geofence.	-20 per instance
3	**Repeated Geofence Breach by Same Drone**	If the same drone breaches the geofence more than once during the mission.	Additional -20 per drone
4	**Manual Input or Reset**	The operator provides manual navigation input, flight control input, waypoint change input, payload drop input, survivor tagging input, system reset, or mission reset after mission start, other than safety abort or emergency recall.	-50 per instance
5	**Crash**	For each instance where any drone crashes during the mission, including uncontrolled ground impact, collision resulting in loss of flight, or crash landing.	-50 per crash
	TOTAL		**600**
Total penalties may be capped at 150 marks, except for safety-critical violations where the organisers may terminate the mission or disqualify the team.			

Mission 2 - NIDAR AirMouse

Phase 4A - Design Review Presentation		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	10
2	Overall Solution Architecture & Indoor Autonomy Strategy	25
3	Technical Approach, Drone Design & Indigenisation Strategy	20
4	GPS-Denied Navigation & Confined-Space Flight	30
5	Indoor Mapping Quality & Mission Planner Display	30
6	Survivor Detection, Localisation & Tagging Accuracy	35
7	Autonomous Mission Execution & Time Efficiency	20
8	Ground Control Station, Mission Map & Survivor Status Reporting	15
9	Fail-Safe Features & Safety	15
	TOTAL	**200**

Phase 4B - Business Strategy Presentation		
No.	**Parameter & Description**	**Max Score**
1	Problem Understanding & Real-World Relevance	30
2	Target Users, Customers & Beneficiaries	20
3	Market Sizing & Deployment Potential	30
4	Business Model & Revenue Approach	30
5	Expenditure Breakdown & Resource Planning	20
6	Funds Raised, Sponsorships & Resource Mobilisation	20
7	Competitive Advantage & Differentiation	20
8	Go-to-Market Strategy & Partnership	20
9	Regulatory, Safety & Adoption Readiness	10
	TOTAL	**200**

Phase 4C - Pre-Flight Inspection		
No.	**Parameter & Description**	**Result**
1	Pre-flight Inspection shall only be a deciding phase, and successful completion of this phase will allow team members to participate in the Final Mission. Teams shall only be permitted to retry this phase once.	Pass or Fail

Phase 4D - Final Mission - Scoring			
No.	**Criteria**	**Scoring Method**	**Max Score**
1	**Survivor Detection & Localisation on 2D Map**	40 marks for each correctly detected survivor whose location is also correctly tagged on the generated 2D map. Location tagging shall mean marking the survivor location with the correct grid reference on the 1 m x 1 m reference grid. Maximum 6 survivors.	240
2	**2D Grid Mapping Accuracy**	Marks awarded based on the number of correctly mapped 1 m x 1 m grid cells within the 15 m x 15 m grid zone, verified against the organiser's reference grid layout. For each grid cell, the team must correctly map the presence of walls, openings, corridors or accessible directions adjoining that grid cell.	220
3	**GCS / Mission Planner Map Display**	Yes/No. Full marks if the GCS displays the generated 2D map, estimated drone position, survivor tags and mission progress. Zero if not achieved.	50
4	**Autonomous Exit from Designated Exit Point**	Yes/No. Full marks if the drone exits from the designated exit point without manual intervention. Zero if not achieved.	50
5	**Fast Completion Bonus**	Yes/No. Full marks if the mission is completed within half of the maximum permitted mission time. Zero if not achieved.	25
6	**Safe Mission Completion**	Yes/No. Full marks if the mission is completed without crash, unsafe contact, uncontrolled landing, or organiser intervention. Zero if not achieved.	15
	TOTAL		**600**

Phase 4D - Final Mission - Penalties			
No.	**Penalty Condition**	**Penalty Description**	**Max Score**
1	**Landing outside the designated launch/landing zone**	For each drone that lands outside the designated landing zone after mission completion, except in an organiser-approved emergency landing.	-10 per drone
2	**Geofence / Mission Boundary Breach**	For each instance in which any drone crosses outside the assigned indoor mission boundary/geofence/permitted maze area.	-20 per instance
3	**Repeated Geofence Breach by Same Drone**	If the same drone breaches the geofence / permitted maze area more than once during the mission.	Additional -20 per drone
4	**Manual Input or Reset**	The operator provides manual navigation input, flight control input, path correction input, survivor tagging input, map correction input, and system or mission reset after mission start, other than safety abort or emergency recall.	-50 per instance
5	**Crash**	For each instance where any drone crashes during the mission, including uncontrolled ground impact, collision resulting in loss of flight, or crash landing.	-50 per crash
Total penalties may be capped at 150 marks, except for safety-critical violations, for which the organisers may terminate the mission or disqualify the team.			

Mission 3 - NIDAR VEGAPilot

Phase 2B - Online Presentation & Shortlisting (Top 100 Team Selection)		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	15
2	Prior Experience & Demonstrated Capability	15
3	Understanding of the Problem Statement	15
4	Preliminary Technical Approach	20
5	Design Tools, Development Tools & Engineering Methods	10
6	Indigenous Component & Non-Chinese Supply Chain Approach	10

7	Development Roadmap, Timelines & Feasibility	15
	TOTAL	**100**

Phase 4A - Design Review Presentation		
No.	**Parameter & Description**	**Max Score**
1	Introduction, Team Composition & Domain Diversity	10
2	Overall Solution Architecture & Aries v3 Integration Strategy	20
3	Autopilot Board Design & Hardware Integration	20
4	Sensor Integration, Navigation & Flight Control	30
5	Software, Firmware & Flight-Control Stack	25
6	Mission Planner & Ground Control Station Integration	20
7	Flight Capability & System Demonstration Strategy	20
8	Indigenous Components, BOM & Supply Chain Compliance	20
9	Safety, Failsafe, Testing & Validation Strategy	35
	TOTAL	**200**

Phase 4B - Business Strategy Presentation		
No.	**Parameter & Description**	**Max Score**
1	Problem Understanding & Real-World Relevance	30
2	Target Users, Customers & Beneficiaries	20
3	Market Sizing & Deployment Potential	30
4	Business Model & Revenue Approach	30
5	Expenditure Breakdown & Resource Planning	20
6	Funds Raised, Sponsorships & Resource Mobilisation	20
7	Competitive Advantage & Differentiation	20
8	Go-to-Market Strategy & Partnership	20

9	Regulatory, Safety & Adoption Readiness	10
	TOTAL	**200**

Phase 4C - Pre-Flight Inspection		
No.	**Parameter & Description**	**Result**
1	Pre-flight Inspection shall only be a deciding phase, and successful completion of this phase will allow team members to participate in the Final Mission. Teams shall only be permitted to retry this phase once.	Pass or Fail

Phase 4D - Final Mission - Scoring			
No.	**Criteria**	**Scoring Method**	**Max Score**
1	**Manual Flight and Flight Stabilisation**	a. Controlled flight response: 30 marks each for successfully demonstrating: (i) controlled roll and pitch response; and (ii) controlled yaw response. b. Stable hover: 20 marks for maintaining a stable hover for a minimum of 30 seconds.	80
2	**Assisted Flight Modes**	50 marks each for successfully demonstrating: (i) Position Hold within ±3 m for a minimum of 30 seconds; and (ii) Altitude Hold within ±1 m for a minimum of 30 seconds.	100
3	**Autonomous Waypoint Mission**	40 marks for each organiser-defined waypoint, up to a maximum of five waypoints, successfully reached and held for a minimum of five seconds before the aircraft automatically proceeds to the next waypoint. The mission shall be executed without manual intervention after initiation. Any manual intervention shall terminate scoring for all remaining waypoints.	200
4	**Return-to-Launch (RTL)**	120 marks if the autopilot successfully initiates Return-to-Launch and completes an autonomous landing. 80 marks if Return-to-Launch is successfully completed, but manual intervention is required only for the final landing due to safety considerations. No marks shall be awarded if Return-to-Launch is not successfully completed.	120
5	**Mission Planner / Ground Control Station Integration**	20 marks each for successfully demonstrating: (i) vehicle connection; (ii) mission upload; (iii) live telemetry; (iv) battery-status display; and (v) mission-progress display.	100
	TOTAL		**600**

Phase 4D - Final Mission - Penalties			
No.	**Penalty Condition**	**Penalty Description**	**Max Score**
1	**Position Hold exceeds ±3 m tolerance**	For each instance in which the aircraft fails to remain within ±3 m of the commanded position during the required 30-second hold period.	−10 per instance
2	**Altitude Hold exceeds ±1 m tolerance**	For each instance in which the aircraft fails to remain within ±1 m of the commanded altitude during the required 30-second hold period.	−10 per instance
3	**Landing outside the designated landing area**	Where the aircraft completes the demonstration but comes to rest outside the organiser-designated landing area, except in an organiser-approved emergency landing.	−20 per instance
4	**Obstacle Contact**	For each instance in which the aircraft, propeller guard, or any attached component contacts an obstacle or course structure.	−10 per instance
5	**Course Boundary Breach**	For each instance in which the aircraft crosses the organiser-defined flight-course boundary.	−20 per instance
6	**Organiser Intervention**	Where the organiser or Flight Inspector is required to intervene to prevent an unsafe condition or recover the aircraft.	−50 per instance
7	**Crash or Uncontrolled Landing**	For each crash, uncontrolled ground impact, collision resulting in loss of controlled flight or crash landing.	−50 per instance
Total penalties may be capped at 150 marks, except in cases involving unsafe operation, deliberate rule violations or safety-critical incidents, for which the organising team may terminate the demonstration or disqualify the team.			

Tie Breaking

9.1. For awards based solely on the Design Review or Business Strategy, , the team with the higher Final Mission score shall be ranked higher.

9.2. For Overall Awards, including Gold, Silver, and Bronze, the team with the higher Final Mission score shall be ranked higher.

9.3. If the tie persists, the team with the lower mission completion time in the Final Mission shall be ranked higher.

9.4. If a tie remains, the Organising Team shall determine the final ranking based on overall technical merit and mission performance.

10. Code of Conduct & Dispute Resolution

Professional Conduct

10.1. All participants, faculty mentors, judges, volunteers and organisers shall conduct themselves professionally and respectfully throughout the competition.

10.2. Participants shall not engage in any behaviour that disrupts the competition, endangers others, damages property, or brings the competition into disrepute.

10.3. Any form of harassment, discrimination, intimidation, misconduct or abusive behaviour shall not be tolerated and may result in disqualification.

Safety Compliance

10.4. Teams shall be solely responsible for ensuring the safety, reliability and airworthiness of their systems.

10.5. Teams shall comply with all safety instructions, inspection requirements and operational guidelines issued by the organisers.

10.6. The organisers may prohibit the operation of any drone, autopilot, communication system or equipment deemed unsafe.

10.7. Failure to comply with safety requirements may result in penalties, mission termination or disqualification.

Academic Integrity

10.8. Teams shall submit only original work developed by their team members.

10.9. Teams may utilise commercially available components, open-source software, and publicly available resources, provided that such use complies with applicable licences and competition rules.

10.10. Plagiarism, misrepresentation, falsification of technical information, fraudulent claims, or deliberate concealment of facts shall constitute a violation of competition integrity.

10.11. Teams found to have copied, replicated or substantially reused another team's work without authorisation may be disqualified.

Organiser Authority

10.12. The Organising Team shall have the overall authority to create, interpret, modify and amend the competition rules, guidelines, timelines, evaluation criteria, awards and other competition requirements from time to time.

10.13. The Organising Team reserves the right to inspect, evaluate and verify any information, document, software, hardware or component submitted by participating teams.

10.14. The Organising Team shall have the final authority to determine eligibility, impose penalties, approve mission participation and declare competition results.

10.15. The Organising Team reserves the right to modify the competition schedule, mission requirements, awards, prizes or operational procedures whenever necessary in the interest of safety, fairness or successful conduct of the competition.

Dispute Resolution

10.16. Participants may raise concerns or seek clarification regarding competition-related matters through the designated communication channels established by the organisers.

10.17. The Organising Team shall make reasonable efforts to review and resolve concerns in a fair and timely manner.

10.18. Decisions of the jury, inspectors and Organising Team regarding technical evaluations, inspections, scoring, penalties, rankings and awards shall be final and binding on all participants.

10.19. Participation in the competition shall constitute acceptance of all rules, guidelines and decisions issued by the Organising Team.

ABOUT DFI

The Drone Federation India (DFI) is a non-profit industry body pioneering the vision of making India a global drone manufacturing and export hub. We serve as an apex body that represents over 550+ drone companies and 5500+ drone professionals across the entire drone ecosystem.

We promote ease of doing business for drone companies, widespread adoption of drone technology across all sectors of the economy and build a skilled community of drone professionals to deliver high-quality world-class products from India.

We bring together government, industry, academia, experts and civil society to build a safe and secure unmanned aerial ecosystem in India.

ABOUT SWAYAAN

The Ministry of Electronics and Information Technology (MeitY) has launched 'SwaYaan – Capacity Building for Human Resource Development in Unmanned Aircraft Systems' in September 2022 with the goal of fostering a national industry-academia collaborative ecosystem for Drone/UAS & related technologies.

The initiative aims to train as many professionals in the UAS sector by 2027 through 1,500+ activities nationwide encompassing five major technical areas towards making India the Global Drone Hub by 2030.

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